# CALEB JONES SHIBU

*Machine Learning & Signal Processing Engineer · Founding Engineer, PAGE Technologies*

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## PROFESSIONAL SUMMARY

Machine-learning engineer specializing in signal processing — building AI systems that turn noisy, real-world sensor data into reliable, deployable intelligence. Six-plus years developing deep-learning methods for complex biomedical and physiological signals (fNIRS/EEG brain–computer interfaces) at SCTIMST, Pontificia Universidad Católica de Chile, the University of Arizona, and the Allen Institute for Neural Dynamics — the same signal-processing expertise he now applies to U.S. climate and environmental technology. As Founding Engineer at PAGE Technologies, architected the firmware, cloud, and machine-learning stack behind an electrochemical nitrate-sensing platform that won a competitive $400,000 EPA SBIR Phase II award and is the subject of a U.S. patent application (filed December 2025) on which he is a named co-inventor. First author of peer-reviewed publications, NeurIPS 2023 co-author, invited journal peer reviewer, Frontiers Research-Topic Editor, IEEE Senior Member, and Full Member of Sigma Xi.

## AREAS OF EXPERTISE

**Machine Learning & Deep Learning:** CNNs, LSTMs, transfer learning, GANs, explainable AI (DeepSHAP, Grad-CAM, LIME), model calibration, sensor fusion, LLM/RAG systems

**Biomedical & Sensor Signal Processing:** fNIRS, EEG, eye-tracking, and electrochemical sensor signals — artifact filtering, feature engineering, real-time classification

**Embedded / IoT & Cloud:** C firmware, MQTT, AWS IoT Core, end-to-end ETL pipelines, real-time streaming (LSL)

**Scientific Software & MLOps:** Python, PyTorch, TensorFlow, OpenCV, PyQt5, scikit-learn, MLflow, Nextflow, GCP, AWS

## SELECTED HONORS, RECOGNITION & SCHOLARLY SERVICE

- **EPA SBIR Phase II Award ($400,000, 2026) —** technical lead and named co-inventor of the funded nitrate-sensing water-quality platform, PAGE Technologies.
- **Named co-inventor, U.S. patent application (filed Dec 2025) —** "Field-Deployable Automated Sampling, Conditioning and Measurement System for Environmental Monitoring," PAGE Technologies.
- **Senior Member, IEEE —** Institute of Electrical and Electronics Engineers, an elevated grade held by fewer than 10% of members (elected 2026).
- **Full Member, Sigma Xi —** The Scientific Research Honor Society (elected 2026).
- **Selected 1 of ~300 applicants —** Software Engineer II, Allen Institute for Neural Dynamics (2024).
- **Co-author, NeurIPS 2023** (Datasets & Benchmarks Track), "The ToMCAT Dataset."
- **Invited Peer Reviewer,** Journal of Neural Engineering (IOP Publishing), 2026.
- **Topic Editor, Frontiers** Research Topic "Computational Approaches to Brain–Computer Interface Development," 2026.
- **Fully Funded Graduate Research Assistantship —** M.S. in Computer Science, University of Arizona.

## PROFESSIONAL EXPERIENCE

**PAGE Technologies, Inc.** *Boulder, CO*

*Founding Engineer (Machine Learning & Signal Processing)* *Sep 2025 – Present*

- Sole founding software/AI engineer; owns the full stack that converts the company's printed electrochemical sensors into a deployable, commercial environmental-monitoring platform.
- Designed the machine-learning calibration and sensor-fusion pipeline that turns raw electrochemical signals into accurate real-time nitrate measurements — the core capability behind the company's $400,000 EPA SBIR Phase II award.
- Engineered embedded C firmware with MQTT-based AWS IoT Core messaging for secure device-to-cloud telemetry, and architected the end-to-end cloud data and analytics infrastructure.
- Named co-inventor on the company's U.S. patent application (filed Dec 2025) for the field-deployable environmental-monitoring platform; established the company's data/ML engineering practice (repositories, model evaluation, field-deployment workflows) from zero.

**Computer Connection of Wisconsin (CCWIS)** *Reedsburg, WI*

*Machine Learning / LLM Engineer* *Jul 2025 – Sep 2025*

- Designed and deployed a retrieval-augmented-generation (RAG) customer-support chatbot (GPT-based) with human-agent escalation and ticket-based support; maintained the company's production web applications.

**Allen Institute for Neural Dynamics** *Seattle, WA*

*Software Engineer II (Machine Learning)* *Mar 2024 – Mar 2025*

- Selected from a pool of roughly 300 applicants for this single Software Engineer II position on the Single-Cell Computation team.
- Retrained the Cellpose neuron-segmentation model on SLAP2 voltage-imaging data via PyTorch transfer learning, enabling automated real-time voltage-imaging analysis used across the lab.
- Built an iGluSnFR motion-correction pipeline that outperformed industry-standard tools (Suite2p, CaImAn, PatchWarp) under extreme motion; contributed a merged upstream fix to the open-source Suite2p tool (MouseLand/suite2p #1122) restoring Python 3.11+ compatibility.
- Developed simulation and OpenCV super-resolution source-extraction pipelines for ground-truth validation and synapse identification; co-authored an image-processing methods paper.

**Biotronics** *Ames, IA*

*Machine Learning Engineer* *Jan 2024 – Feb 2024*

- Enhanced ResNet-50 interpretability for livestock-fat prediction using SHAP and Grad-CAM to quantify feature importance and build stakeholder trust.

**University of Arizona — School of Information & Dept. of Computer Science** *Tucson, AZ*

*Graduate Research Assistant — ToMCAT Project* *Aug 2021 – Dec 2023*

- Sole fNIRS specialist on a federally supported AI teaming project; architected the multimodal (fNIRS/EEG/eye-tracking) data-acquisition infrastructure and collected high-fidelity data from 40 three-person teams (approximately 120 participants).
- Developed the core ML pipelines and real-time brain-state prediction models and built the ToMCAT-offline-Viz PyQt5 visualization tool; co-authored "The ToMCAT Dataset," NeurIPS 2023.

**Sree Chitra Tirunal Institute for Medical Sciences & Technology (SCTIMST)** *Kerala, India*

*Project Scientist* *Jan 2021 – Jul 2021*

*Research Intern* *Sep 2019 – Sep 2020*

- First at the institute to apply modern deep learning to fNIRS-BCI: sliding-window CNN/LSTM models raised classification accuracy from ~55% to 97%; integrated into a real-time PyGame neurofeedback interface.
- First-authored the resulting IEEE ICIIBMS 2020 paper; promoted from Research Intern to full-time Project Scientist within an Institution of National Importance.

**Pontificia Universidad Católica de Chile (Sitaram Lab)** *Santiago, Chile (Remote)*

*Research Intern — Explainable AI for BCI* *Sep 2020 – Jan 2021*

- Built an explainable-AI fNIRS classifier (DeepSHAP) that reached 97% accuracy distinguishing active vs. passive brain states in stroke-related motor preparation, mapping predictions to specific cortical channels; first-authored the resulting article in Frontiers in Human Neuroscience (2023).

## SELECTED PUBLICATIONS


Shibu, C. J., Sreedharan, S., Arun K.M., Kesavadas, C. "Comparison of classification performance of handpicked, handcrafted, and automated-features for fNIRS-BCI system." 2020 IEEE ICIIBMS, pp. 152–157. (First author)

Shibu, C. J., Sreedharan, S., Arun K.M., Kesavadas, C., Sitaram, R. "Explainable artificial intelligence model to predict brain states from fNIRS signals." Frontiers in Human Neuroscience, 2023. (First author)

Pyarelal, A., Duong, E., Shibu, C. J., et al. "The ToMCAT Dataset." NeurIPS 2023, Datasets & Benchmarks Track.

Shibu, C. J. "Decoding Emotional Responses: A Comparative Study of fNIRS and EEG Neuroimaging Techniques." M.S. Thesis, University of Arizona, 2023.

Xie, M. E., Friedrich, J., Wirsching, E., Shibu, C. J., et al. "GIAnT: a Glutamate Imaging Analysis Toolbox." bioRxiv preprint, 2026. doi:10.64898/2026.08.07.743580


## EDUCATION

**M.S., Computer Science — University of Arizona** Fully Funded Graduate Research Assistantship · Aug 2021 – Dec 2023

**Bachelor of Engineering in Computer Engineering (First Class) — Padre Conceição College of Engineering, Goa University, India** Aug 2014 – May 2018